

Knowing a Fact vs. Being Able to Act

Separating declarative knowledge gaps from procedural competence gaps in a taught reasoning system

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ABSTRACT

A system that learns by being taught accumulates two distinct kinds of mastery over any subject: *declarative knowledge* (the facts it holds) and *procedural competence*: the operations it can actually carry out. A system that conflates the two misreads its own state, mistaking “I have not yet been told this fact” for “I am not competent in this area,” or the reverse. We formalize the two as independent, separately measured axes, state five falsifiable hypotheses about whether a taught reasoning system can discriminate between them and whether the axes stay orthogonal, and test them on a running symbolic system. Teaching three connected facts about a subject raised its declarative coverage ($0.229 \rightarrow 0.293$) while leaving procedural competence exactly unchanged ($0.500 \rightarrow 0.500$). Three hypotheses held immediately. A fourth (detecting a *specific* missing fact inside an otherwise well-developed subject) required a component the framework predicted but the implementation had not yet supplied: a detector to populate the register of acknowledged ignorance. With that detector in place all sixteen checks passed, and negative controls confirm it neither fabricates gaps where none exist nor disturbs the competence estimate. We contrast this behavior with an autoregressive language model, which represents no such boundary and therefore cannot abstain on the basis of it.

1 Two kinds of not-knowing

Over any subject, a learning system can be strong along two different dimensions that are easy to confuse. *Declarative knowledge* is how much it knows: the facts it has been told and connected together. *Procedural competence* is what it can do: the operations it can actually execute to reach a goal. The distinction is old in

the study of mind: knowing *that* versus knowing *how* [1], and, in cognitive architecture, declarative versus procedural memory [2].

Declarative knowledge and procedural competence are separable, independently measurable properties of a taught system. Representing them as orthogonal axes lets the system diagnose which kind of gap it faces (a missing fact to be acquired, or a missing operation to be practiced) and abstain honestly on that basis.

2 Why conflating them fails

A missing fact and a missing operation call for opposite responses. A *knowledge gap* is closed by acquisition: being taught the fact, or looking it up. A *competence gap* is closed by practice or derivation: searching for an operation that produces the goal. Routing one to the other wastes effort and corrupts the system's estimate of what it can do: treat “I was never told this one fact” as “I am not competent in this whole area” and the system needlessly discounts real ability; treat “I cannot perform this operation” as “I just lack a fact” and it will keep asking to be told things no fact can supply.

An autoregressive language model has a single channel (next-token likelihood) and no internal variable separating “I hold little information about this” from “I cannot perform this operation.” Both surface identically, as a distribution over tokens. Part of what is called hallucination is precisely the absence of this boundary: without a representation of *this subject's coverage* as distinct from *this operation's feasibility*, the model cannot know it is missing something, so it answers fluently anyway. Selective-prediction methods add an abstention head over likelihood [3]; they do not add the two-axis boundary studied here.

3 Method: two axes, two producers

Declarative coverage is a maturity measure over the graph of taught concepts, rising as connected facts are taught. It is a normalized score in [0,1] combining how many concepts the subject contains (size), how densely they interconnect (density), and how many distinct kinds of relation link them (variety), weighted 0.4 / 0.3 / 0.3. **Procedural competence** is a probabilistic belief that the system has learned the subject's operations; it moves only on the outcomes of attempting operations, never on being taught facts. The load-bearing property is that *neither producer writes the other's variable*.

Deciding which gap is faced. When a goal cannot be achieved, a structural diagnosis walks the chain from symbol to action and returns exactly one typed deficit. Two are central: an *unrepresented-predicate gap* (the goal concept appears nowhere in the subject's vocabulary, so the response is to acquire a new concept) and a *missing-operation gap* (the concept is represented, but nothing the system knows produces it, so the response

is to explore for an operation). Crucially, diagnosing a gap does not by itself change the competence belief: a diagnosis is not evidence of incompetence.

The declarative twin. The system carries a first-class representation of *acknowledged ignorance*: a subject-scoped question it knows it cannot answer, with an information-value and a resolution strategy. Our core addition is a detector that, for a missing relation on a *known, in-subject* concept, registers exactly such an entry (while leaving the competence belief untouched) and refuses to register anything when the subject is out of scope or the relation is already present. A registered gap is surfaced as an exploration target and converted into an *acquisition* goal (be taught, or look up), not an experiment to build an operation.

4 Experiment

To measure the axes cleanly, a mature subject was built by teaching a small, connected, deliberately fictional taxonomy through the ordinary teaching interface (fictional terms keep the test subject isolated and deterministic). Six facts built the subject; three more connected facts were taught for the orthogonality test. Five hypotheses were then evaluated live: discrimination (H1), routing (H2), axis orthogonality (H3), declarative-gap detection with negative controls (H4), and gap pursuit (H5).

5 Results

5.1 The axes are orthogonal

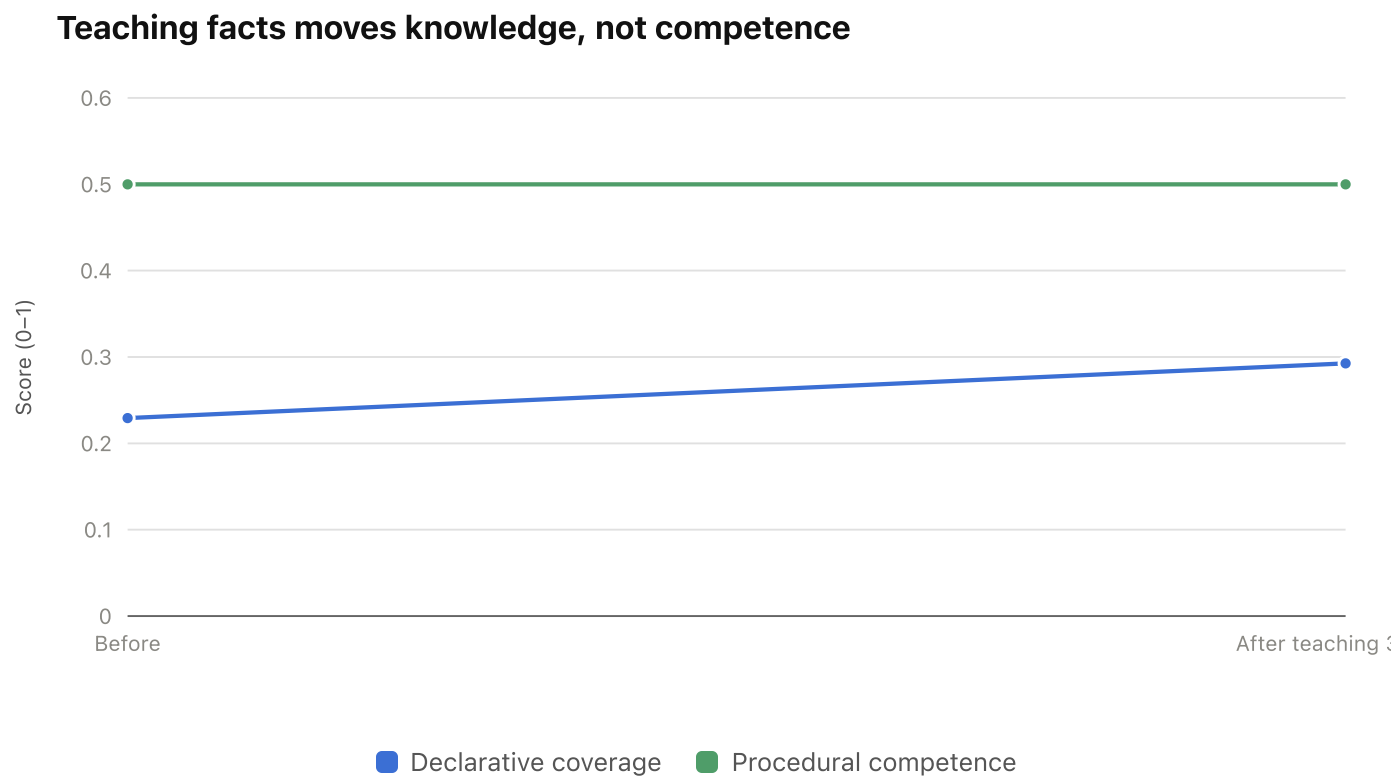


Figure 1. After teaching three additional connected facts, declarative coverage rose from 0.229 to 0.293, while procedural competence held at exactly 0.500 (change below 10^{-9}). One axis moves; the other does not: the orthogonality the design requires.

5.2 Discrimination, routing, detection, pursuit

All sixteen checks passed

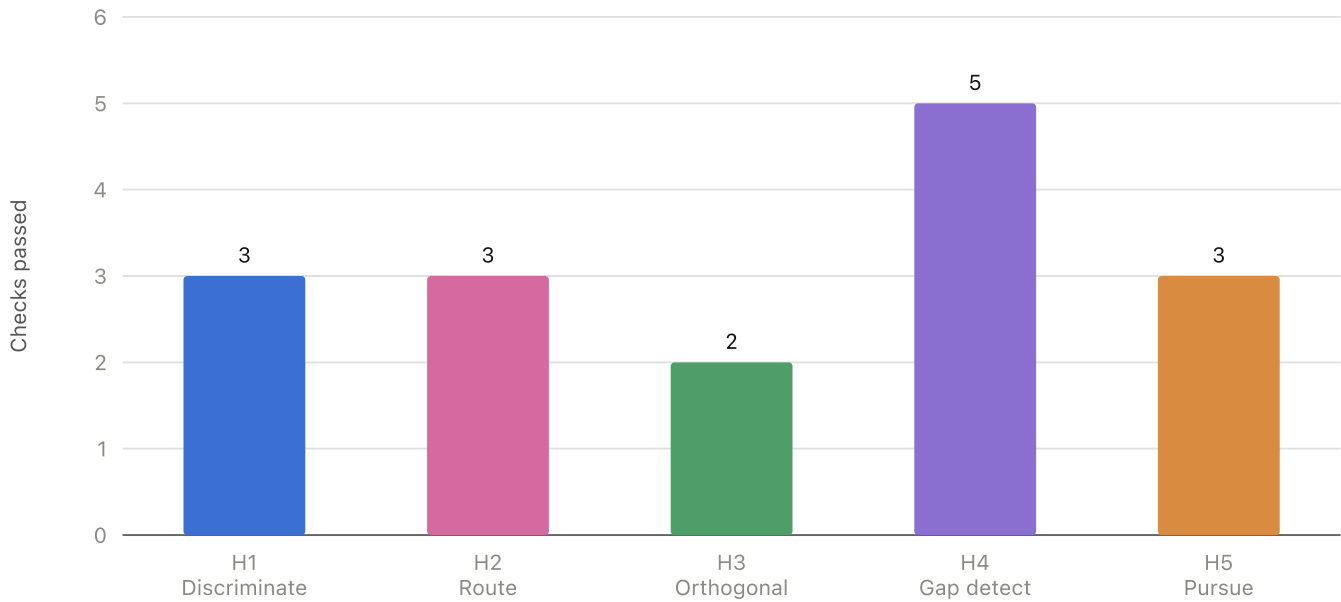


Figure 2. Checks passed by hypothesis: discrimination 3/3, routing 3/3, orthogonality 2/2, declarative-gap detection 5/5, gap pursuit 3/3, 16/16 in total. Discrimination returns the correct gap type (0 actionable operations for a missing-operation gap; predicate-not-in-vocabulary for an unrepresented-predicate gap); routing sends the two to acquire-a-concept versus explore-for-an-operation.

5.3 A gap it detects, without fabricating gaps

Detecting a specific missing fact inside a mature subject isolates the one piece the two-axis framework calls for that the implementation had not yet supplied: the register of acknowledged ignorance was defined, but no producer wrote to it, so the register stayed empty and the gap went unregistered. That is the discriminating measurement; it locates the missing producer precisely, rather than attributing the miss to the axes themselves. With the detector added, a missing relation on a known concept registered exactly one acknowledged-unknown (count $0 \rightarrow 1$, information-value 0.6), which became one exploration target (entropy 0.94) routed to acquisition, while competence stayed at 0.500. Two negative controls confirm specificity.

Situation	Gap registered?	Competence
In-subject, relation not yet held	yes ($0 \rightarrow 1$)	0.500 (unchanged)
Out-of-scope subject	no	0.500 (unchanged)
Relation already present	no	0.500 (unchanged)

Table 1. Gap detection and its negative controls. The detector fires only for an in-subject, not-yet-held relation, and never disturbs the competence estimate.

6 A third kind of not-knowing

The two axes classify a deficit as a missing fact or a missing operation, and route each to its own remedy. Work since this study identifies a third case that is neither, and that the framework’s own logic requires be told apart from both: the concept is represented, an operation is available, and what the system lacks is a *discriminating example*. Its evidence admits more than one hypothesis, and nothing in that evidence prefers any of them.

The three are distinguished by signature, not by degree (Table 2). A missing fact shows up as a concept absent from the subject’s vocabulary, and is closed by acquisition. A missing operation shows up as a represented concept with no producer, and is closed by practice or search. Undetermined evidence shows up as several surviving hypotheses that agree on some instances and disagree on others, and is closed by neither: no amount of practice narrows it, and acquiring an arbitrary further fact usually does not either. What closes it is one specific case, and the system can say which.

Deficit	Signature	Correct response
Missing fact	concept not in the subject’s vocabulary	acquire the concept
Missing operation	concept represented, no producer for it	practise or search for an operation
Undetermined evidence	several hypotheses survive and disagree	obtain the one case that separates them

Table 2. Three typed deficits. The first two are the axes this paper measures; the third is distinguished from both by what closes it.

The behaviour under the third deficit has been measured on the same substrate. Across 32 inductions, teaching sets whose positive examples shared an incidental feature left more than one hypothesis standing in every case, and teaching sets whose positives varied left exactly one in every case, so the condition is produced by the evidence rather than by the subject. Scored over 288 held-out cases, asserting on any one sur-

viving hypothesis would have produced **26** assertions of cases that were false; asserting only what every surviving hypothesis supports produced **none**, while still asserting 73% of the true cases. The system does not guess its way out of an ambiguity it can detect.

It also states what would end it. An undetermined induction reports the case that would eliminate one of its hypotheses, and supplying that case closed the ambiguity in **sixteen of sixteen** inductions, in a mean of 2.6 rounds, where supplying randomly chosen further examples under the same budget closed **one** of sixteen. This is the same discipline as the register of acknowledged ignorance in §5.3, applied one level up: there the system records a fact it knows it lacks, and here it records an *experiment* it knows it needs. Both are entries in what the system honestly does not know, and both are actionable rather than merely reported.

7 Implementation status

The detector reported in §5.3 is implemented and in service. The register of acknowledged ignorance is written by live producers on both the domain and the autonomous-reasoning paths, entries persist rather than living only for the duration of a run, and resolution is a first-class transition, so an acknowledged unknown that is later answered is closed rather than accumulating. The sixteen checks in §5 run against that implementation.

The before-and-after in §5.3 is retained on purpose. The count $0 \rightarrow 1$ movement, with competence held flat at 0.500 across it, is what shows the detector registers a declarative gap *without* disturbing the procedural axis: the orthogonality claim the paper is making. Reporting only the post-detector state would remove the evidence for it.

8 Limitations and conclusion

The coverage measure is a specific normalized combination of size, density, and variety; other reasonable measures exist, and the absolute numbers are not the point: the separation of the two axes is. The subject is a small fictional taxonomy chosen for determinism; scaling the discrimination to large, real, partially-taught subjects is future work, as is a competence axis defined over many operation types rather than one. The third deficit in §6 is characterised by what closes it and by the system's behaviour under it; whether registering one leaves the competence axis untouched, as registering a declarative gap does, has not yet been measured and is the natural next check. The result we stand on is qualitative and robust: a taught system can hold two independent measures of mastery, decide which kind of gap it faces, and abstain honestly: a boundary a single-channel model does not represent.

References

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